8- QUEENS PROBLEM

Aim :
Write a program to solve the 8- queens problem program.

Code :

```
N = 8
board = [[0]*N for _ in range(N)]
def is_safe(board, row, col):
    for i in range(col):
        if board[row][i] == 1:
            return False
    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1
    i, j = row, col
    while i < N and j >= 0:
        if board[i][j] == 1:
            return False
        j += 1
        i -= 1
    return True
def solve(board, col):
    if col == N:
        return True
    for row in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            if solve(board, col+1):
                return True
            board[row][col] = 0
    return False
```

```
if solve(board, 0):
    for row in board:
        print(*row)
else:
    print("No solution exists")
```

Output:

```
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0
0 0 0 0 0 1 0 0
0 0 1 0 0 0 0 0
```